



Metal-ion-assisted construction of cyano group defects in g-C₃N₄ to simultaneously degrade wastewater and produce hydrogen

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ABSTRACT

Converting inexhaustible solar energy into chemical energy for environmental contaminant control and fossil fuel substitution has been considered a sustainable and promising strategy. In this paper, an easy-to-operate salt-assisted method is used to synthesize a bifunctional graphitic carbon nitride (BaCN-C₃N₄) co-decorated with Ba²⁺ doping and cyano defects. This novel BaCN-C₃N₄ is applied to produce hydrogen from organic wastewater efficiently and stably. Experimental characterization and DFT calculation suggest that Ba²⁺ is trapped in the carbon nitride intra-layer ring cavity through ion-dipole interaction, meanwhile the presence of Ba²⁺ promotes the production of cyano defects by partial heptazine ring opening. The mechanism of the excellent photocatalytic activity of BaCN-C₃N₄ is that the synergistic action of Ba²⁺ and cyano group promotes stronger visible light absorption, narrower band gap and higher photogenerated electron-hole separation. As a consequence, BaCN-C₃N₄ achieves an outstanding hydrogen production rate (10316 μmol/g/h) from wastewater, which is nearly 5.5 times that of the pristine g-C₃N₄. This investigation cleverly used the salt-assisted method to construct cation and cyano defects co-modified carbon nitride, achieving the goal of coinstantaneous pollutant degradation and energy development, and providing a new strategy for semiconductor photocatalysts to convert solar energy.

1. Introduction

Photocatalysis has been recognized as an ideal technology for treating environmental contaminants and producing hydrogen [1–3]. Graphitic carbon nitride (g-C₃N₄), as a metal-free semiconductor photocatalyst, has excellent redox performance due to its suitable conduction band position (ca. −1.3 eV) and valence band position (ca. + 1.4 eV), so that it is especially suitable for the elimination of organic contaminants and the generation of hydrogen [4–6]. Numerous reports pointed out that g-C₃N₄-based photocatalysts can efficiently remove organic pollutants or produce hydrogen in an independent system [7–9].

How to use g-C₃N₄-based catalyst to obtain hydrogen energy from wastewater and simultaneously degrade pollutants is a challenging research. This is based on the fact that the consumption of photo-generated electrons by O₂ will reduce the efficiency of hydrogen production in the process of degrading pollutants, meanwhile photo-generated holes with weak oxidizing ability are not conducive to the efficient degradation of organic pollutants in the process of hydrogen

production [10]. Recently, a few researchers reported that g-C₃N₄-based photocatalyst can generate hydrogen from organic wastewater, which can simultaneously solve environmental pollution and energy shortage issues [11–13]. These research pointed out that the g-C₃N₄-based photocatalyst is an extremely promising catalyst to degrade organic pollutants and develop energy in the same system. However, the synthetic methods in these reports have disadvantages such as low yield, harsh synthetic methods, complicated operations and large consumption of dangerous organic reagents. Therefore, it is urgent to explore a cost-effective, easy-to-operate and non-toxic synthesis method to prepare an excellent g-C₃N₄-based catalyst to achieve the degradation of pollutants and hydrogen production simultaneously in the same system.

Metal-ion-assisted method is widely used to prepare metal ion-doped modified g-C₃N₄. The alkali metal K⁺ [14], Na⁺ [15] and transition metal Mn²⁺ [16], Co²⁺ [17], Cu²⁺ [18] doped in g-C₃N₄ with this method have been demonstrated to improve the photocatalytic activity of g-C₃N₄. Because the g-C₃N₄ nitrogen tank is filled with six nitrogen lone-pair electrons, they can easily establish ion-dipole interactions

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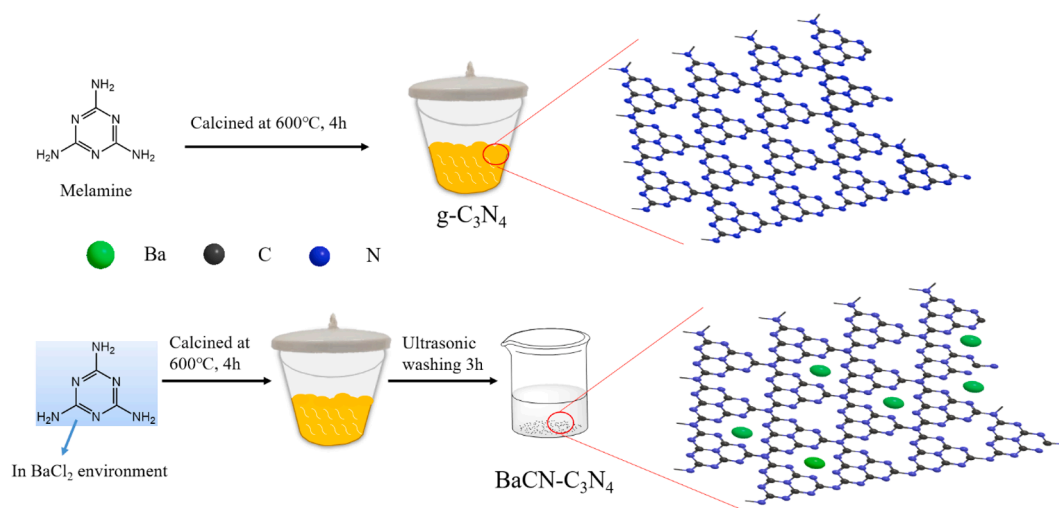
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Scheme 1. Preparation of bifunctional BaCN-C₃N₄ photocatalyst.

with the cations entering the tank for ions doping to optimize its energy band and electronic structure [19]. Moreover, it is interesting that researchers recently revealed that the metal-ion-assisted method can introduce cyano defects in the g-C₃N₄ framework, which could effectively promote the separation of photogenerated carriers due to its strong electron withdrawing effect [20,21]. Obviously, metal-ion-assistance seems to be an ideal and feasible method to introduce donor-type metal ligands and acceptor-type functional groups into the g-C₃N₄ framework to optimize its energy band and electronic structure. But, there is few report on the modification of g-C₃N₄ by cation doping and cyano group simultaneously via an easy-operated method. Ba²⁺, a resource-rich alkaline earth metal cation, has an ion radius of 1.35 Å, which is smaller than the size of the intra-layer ring cavity of g-C₃N₄ (4.72 Å), so that it can be trapped in the heptazine ring cavity through ion-dipole interactions with N around the nitrogen cavity [22,23]. However, the barium-ion-assisted method simultaneously introduces Ba²⁺ doping and cyano group defects into the g-C₃N₄ to improve its photocatalytic activity for pollutants abatement and hydrogen generation at the same time has not been reported.

Based on the above analysis, we reported for the first time a Ba²⁺ doping and cyano defects co-modified g-C₃N₄ photocatalyst (BaCN-C₃N₄) prepared by an alkaline earth metal salt-assisted method. As expected, BaCN-C₃N₄ has narrower band gap, faster photogenerated electron-hole separation, and higher photocatalytic activity for hydrogen production in antibiotic wastewater. To the best of our knowledge, this work first reported that the adjustment of the energy band and electronic structure of g-C₃N₄ by cation doping and cyano defects to improve the photocatalytic performance and accelerate the elimination of organic pollutants and conversion of energy in the same system. We expect that this work could provide a new avenue for the effective conversion of solar energy with g-C₃N₄-based photocatalysts, and improve its application value in the fields of environmental governance and energy development.

2. Experimental section

2.1. Photocatalyst preparation

A bifunctional carbon nitride based photocatalyst (BaCN-C₃N₄) co-modified with Ba²⁺ and cyano group was successfully synthesized by doping Ba²⁺ to construct cyano group defects in g-C₃N₄ framework, using a salt-assisted method. Specifically, 10 g of C₃H₆N₆ and a certain amount of BaCl₂ (0.5 g, 0.7 g and 0.9 g) were placed in a 100 mL ceramic crucible, and 20 mL of deionized water was added and fully stirred. The mixture was then sonicated in an ultrasonic cleaner for 20 min, and

further dried overnight in a blast drying oven to obtain a precursor. Thereafter, the precursor was raised from room temperature to 600 °C in a muffle furnace at a heating rate of 5 °C/min, and maintained at this temperature for 4 h. After cooling to room temperature, the calcined material was fully ground to obtain a powder sample. Further, 1 g of the powder sample was placed in 250 mL of deionized water and ultrasonically washed for 3 h to fully remove water-soluble impurities. Finally, after centrifugation at 5000 r/min, the sample m-BaCN-C₃N₄ was dried and labeled, where m (5%, 7%, and 9%) refers to the relative content of BaCl₂. Pure g-C₃N₄ was prepared by directly calcining 10 g of melamine (without BaCl₂), labeled as g-C₃N₄. The preparation process of BaCN-C₃N₄ is shown in Scheme 1.

2.2. Characterization

The phase structure, functional groups and element composition of the samples were characterized by X-ray diffraction (XRD) (XRD-6100, Shimadzu, Japan), Fourier transform infrared spectroscopy (FT-IR, IRPrestige-21, Shimadzu, Japan) and X-ray photoelectron spectroscopy (XPS, Thermo Scientific, ESCALAB 250xi, USA). The content of the doped Ba²⁺ was determined by inductively coupled plasma optical emission spectrometry (ICP-OES, Spectro genesis, Germany). Transmission Electron Microscopy (TEM, FEI, Tecnai G2F20, USA), N₂ adsorption-desorption apparatus (BET, ASAP 2020, USA) and Energy Dispersive X-ray Spectroscopy (EDS SU1510, Hitachi, Japan) were used to observe the morphology, physical properties and element distribution of the catalyst. The optical properties and catalytic mechanism were studied by UV-visible diffuse reflectance spectroscopy (UV-vis DRS, Shimadzu, Japan), Photoluminescence (PL, F-7000, Hitachi, Japan), Electron spin-resonance spectroscopy (ESR, Hitachi, FA200, Japan), LC-MS/MS with a combination of 1260 Infinity HPLC (Agilent, USA) and a mass spectrometer (6420 Triple Quadrupole LC/MS, Agilent, USA).

2.3. DFT calculations

The calculations of geometry configurations and the band structures were carried out by the Vienna Ab-initio Simulation Package (VASP), taking advantage of the density functional theory (DFT) with the Projected Augmented Wave (PAW) method. The Perdew-Burke-Ernzerhof (PBE) function was employed to illustrate the exchange and correlation effects. SCF convergence criteria was set to be 10⁻⁵ eV in the calculation. Gaussian smearing with 0.1 eV width was adopted in the geometry optimization. The relaxation convergence criteria was set to be 0.03 eV/Å. The cut-off energy was set to be 400 eV. The Γ -centered Monkhorst-Pack grids were set to be 1 × 1 × 1 for performing the

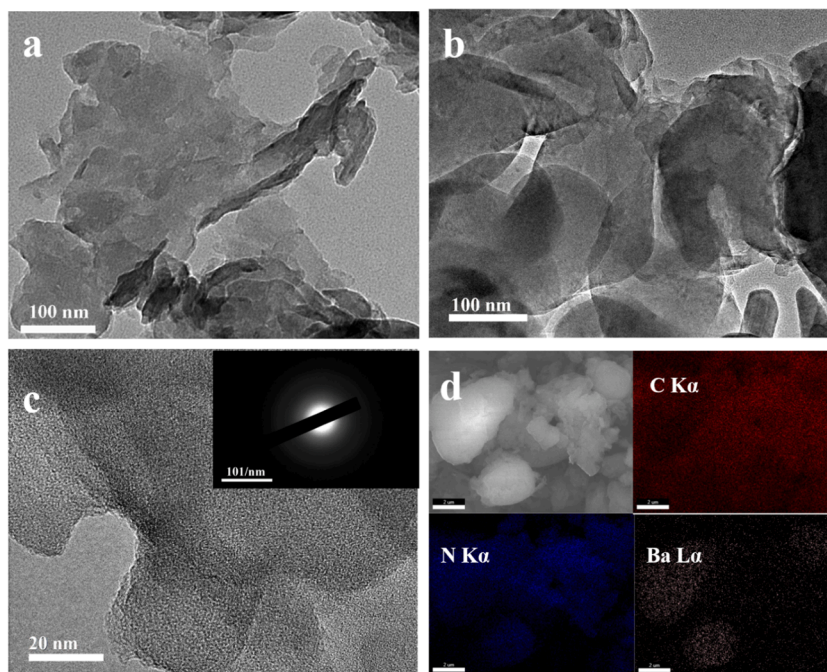


Fig. 1. TEM images of g-C₃N₄ (a) and 7%-BaCN-C₃N₄ (b). HRTEM and SAED (inset) images of 7%-BaCN-C₃N₄ (c); SEM-mapping of 7%-BaCN-C₃N₄ (d).

calculations. As for band structure and electronic configuration calculation, tetrahedron method with Blöchl corrections was adopted. The energy cutoff was set to be 600 eV and a denser k-point mesh of $2 \times 2 \times 1$ was adopted. We firstly constructed a single-layer g-C₃N₄ system (g-C₃N₄). Then, Ba²⁺ was introduced into part of the heptazine ring cavity (Ba-C₃N₄) to study the influence of Ba²⁺ on the CN structure framework and energy band. Lastly, a part of Ba²⁺ doping and cyano (-C≡N) co-modified g-C₃N₄ (BaCN-C₃N₄) was also constructed to reveal their synergism effect on the band structure and electronic configuration of the g-C₃N₄ system. For all the systems, a ~ 20 Å vacuum thickness was added in the z-direction of the simulation box, preventing the interactions between the adjacent surfaces.

2.4. Photocatalytic test

25 mg photocatalyst powder, 50 mL TEOA aqueous solution (45 mL deionized water and 5 mL TEOA) and 2 mL chloroplatinic acid (1 g/L, ~ 3 wt% Pt) jointly constitute the reaction system in a 100 mL gas-closed Pyrex reactor. Subsequently, the reaction system was connected to a glass closed gas circulation system and a continuous circulation cooling tank was used to control the temperature (15 °C). A 300 W Xe lamp matched with a 420 nm cut-off filter ($\lambda > 420$ nm) was used as the light source, and the average light power obtained by CEL-NP2000 spectroradiometer (Education Au-light, China) was about 300 mW/cm². Before the light irradiation, the whole reaction system was evacuated. Finally, the amount of generated hydrogen (GC 7920, Beijing) was analyzed by gas chromatograph. The apparent quantum efficiency (AQE) of the synthesized photocatalyst was obtained by Equation (1):

$$AQE = \frac{2 \times \text{number of hydrogen molecules}}{\text{number of incident photon}} \times 100\% \quad (1)$$

It is important and valuable that the hydrogen evolution and degradation efficiency of BaCN-C₃N₄ were simultaneously investigated in wastewater. Only by replacing the deionized water in the above independent hydrogen evolution system with tetracycline wastewater, the tetracycline was removed while producing hydrogen. The concentration variation of TC was obtained by UV-vis spectrum at the maximum absorption wavelength. In addition, the independent degradation process of TC by BaCN-C₃N₄ was also considered. The exhausted analysis

process is described in Text S1.

2.5. Photoelectrochemical analysis

3 mg of the catalyst was dispersed in a mixed solution of 250 μ L of water, 250 μ L of absolute ethanol, and 10 μ L of Nafion, and the mixed solution was further ultrasonically dispersed for 30 min to obtain a dispersion. This dispersion was uniformly deposited on a 15 cm $\times$ 15 cm ITO conductive glass, then dried under vacuum at 60 °C for 30 min. The photocurrent was estimated in a three-electrode battery connected to electrochemical workstation (CHI660E, Chenhua, China) using glassy carbon as the counter electrode, carbon rod as reference, and Na₂SO₄ aqueous solution (0.2 M) as the electrolyte. The light source was a 300 W Xe lamp. The photocurrent with an on/off cycle was studied with an applied potential of 0.6 V every 20 s.

3. Results and discussion

3.1. Morphology and phase composition

In Fig. 1, the morphology and microstructure of the catalyst are displayed. g-C₃N₄ is shown as a sheet-like stacked structure with an amorphous crystallinity in Fig. 1a [24]. The 7%-BaCN-C₃N₄ has smaller size (Fig. 1b), and no clear lattice fringes are observed in the higher magnification (Fig. 1c). Furthermore, no obvious diffraction ring is observed in the Selected Area Electron Diffraction (SAED) of 7%-BaCN-C₃N₄ (in the inset in Fig. 1c). The above conclusions indicate that 7%-BaCN-C₃N₄ maintains the amorphous sheet structure of g-C₃N₄. The N₂ adsorption-desorption isotherm and the corresponding Barrett-Joyner-Halenda pore size distribution curve are shown in Figure S1. The specific surface area of 7%-BaCN-C₃N₄ (19.05 cm²/g) is considerably larger than that of g-C₃N₄ (14.45 cm²/g), and the pore size distribution curve shows that 7%-BaCN-C₃N₄ has a smaller pore structure (in the range of 2–10 nm). These physical properties indicate that 7%-BaCN-C₃N₄ might have more active sites, which is more conducive to the progress of the photocatalytic reaction. In Fig. 1d, the SEM element mapping of 7%-BaCN-C₃N₄ shows that three elements C, N, and Ba are evenly distributed, which indicates that Ba element was successfully introduced into g-C₃N₄ through high-temperature calcination.

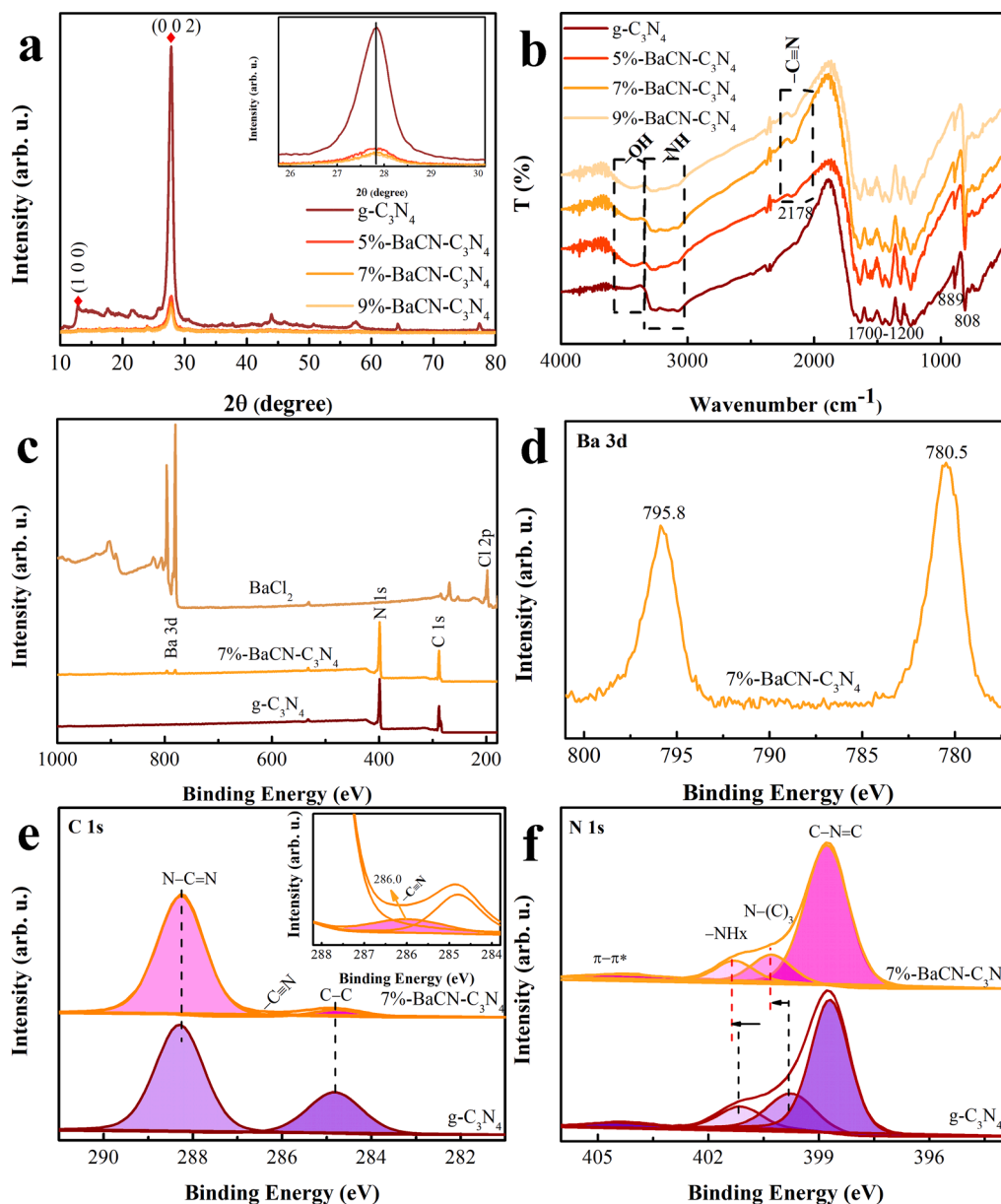


Fig. 2. XRD patterns (a) and FT-IR spectra (b) of g-C₃N₄, m-BaCN-C₃N₄; XPS full spectra (c), narrow spectra Ba 3d (d) and high-resolution spectra C 1 s (e) and N 1 s (f) of catalyst.

The phase structures of g-C₃N₄ and m-BaCN-C₃N₄ catalysts analyzed by XRD pattern are shown in Fig. 2a. The layered g-C₃N₄ exhibits two characteristic diffraction peaks at 12.9° and 27.8°, which are consistent with the (1 0 0) inter-plane stacking and (0 0 2) interlayer stacking motif of the heptazine unit, respectively [25,26]. For m-BaCN-C₃N₄, the main peak positions are still at 12.9° and 27.8°, which indicates that their main structure is g-C₃N₄. Interestingly, the intensity of the (0 0 2) characteristic peak of BaCN-C₃N₄ gradually weakens with the increase of Ba²⁺ doping content, indicating that the polymerization of g-C₃N₄ should be affected by the presence of Ba²⁺. The weakening peaks imply that the long-range ordered atomic arrangement of g-C₃N₄ is destroyed and there might be some defects (i.e., cyano group) [27,28]. In Fig. 2b, the functional groups of g-C₃N₄ and m-BaCN-C₃N₄ are obtained by FT-IR spectroscopy. For pristine g-C₃N₄, the strong band peaks at 1700–1200 cm⁻¹ are related to the stretching vibration of N–C≡N heterocycles, while the vibration peak at 808 cm⁻¹ is assigned to the s-triazine ring [29,30]. The peaks at ~3400 cm⁻¹ and ~3200 cm⁻¹ are derived from the N–H and O–H of the g-C₃N₄ structure, respectively. However, a new

characteristic peak at 2178 cm⁻¹ in m-BaCN-C₃N₄ is attributed to the stretching vibration of the cyano group (–C≡N) [31,32]. This confirms that the cyano group (strong electron withdrawing group) is successfully introduced into the g-C₃N₄ structure through the metal-ion assisted method.

The XPS full spectra in Fig. 2c demonstrate that Ba is successfully doped into the g-C₃N₄ by the Ba-ion assisted method. Furthermore, Fig. 2d shows the Ba 3d narrow spectra of 7%-BaCN-C₃N₄. It is detected that the peaks at binding energies of 780.5 eV and 795.8 eV are assigned to Ba 3d5/2 and Ba 3d3/2, respectively, which indicate that Ba element exists in 7%-BaCN-C₃N₄ as Ba²⁺ [33]. The amount of Ba²⁺ doping in 5%-BaCN-C₃N₄, 7%-BaCN-C₃N₄ and 9%-BaCN-C₃N₄ are obtained by ICP-OEC, which are 2.44 wt%, 2.69 wt% and 2.80 wt%, respectively (Figure S2). The binding energy of 284.80 eV and 288.30 eV are attributed to the C–C bonds (instrument) or amorphous carbon and N–C≡N in the tris-triazine structure (in Fig. 2e) [34,35]. There are four characteristic peaks in Fig. 2f, namely 398.70 eV, 399.76 eV, 401.13 eV and 404.58 eV, which correspond to C–N=C vibration, N–C₃, –NHx and

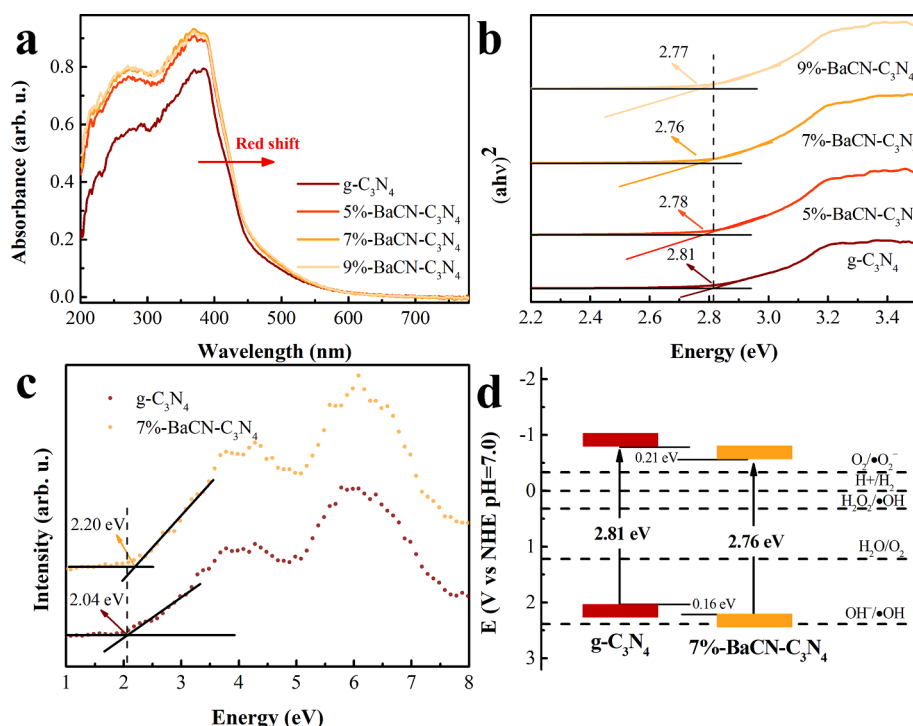


Fig. 3. UV-vis diffuse reflectance spectra (a) and Tauc plots (b) of g-C₃N₄ and m-BaCN-C₃N₄. VB-XPS spectra (c) and Band structure (d) of g-C₃N₄ and 7%-BaCN-C₃N₄.

$\pi-\pi^*$, respectively [36]. Compared with g-C₃N₄, a characteristic peak (286.5 eV) attributed to cyano group is found in C 1 s of 7%-BaCN-C₃N₄, which once again prove that the partial structure of g-C₃N₄ is influenced by Ba²⁺ to construct cyano group [32]. The characteristic peak of N 1 s in 7%-BaCN-C₃N₄ shifts to higher binding energy compared to g-C₃N₄, which may be due to the chemical coordination between the doped Ba²⁺ and g-C₃N₄ aromatic ring [37]. In 7%-BaCN-C₃N₄, the peak area of N-C₃ decreases but -NHx increases. It could be speculated that the doped Ba²⁺ destroys the long-range ordered structure of g-C₃N₄, leading to the breaking of part of the bridge N to form cyano group.

3.2. Optical properties and band structures

The excellent visible-light absorption is closely related to the outstanding photocatalytic activity of the photocatalyst, so the light absorption range of the synthesized catalyst is recorded (in Fig. 3). For pristine g-C₃N₄, it has a light absorption band edge at ~ 460 nm, corresponding to previous reports [38,39]. Interestingly, the series BaCN-C₃N₄ has a red-shifted light absorption band edge (approximately at ~ 470 nm), which means they have a wider range of visible light adsorption. The band gap of the synthesized sample was obtained by Kubelka-Munk formula (Fig. 3b)[6,40]. The band gaps of g-C₃N₄, 5%-BaCN-C₃N₄, 7%-BaCN-C₃N₄ and 9%-BaCN-C₃N₄ are 2.81 eV, 2.78 eV, 2.76 eV and 2.77 eV, respectively. Furthermore, the VB-XPS spectra provide the valence band positions of g-C₃N₄ and 7%-BaCN-C₃N₄, which are 2.04 eV and 2.20 eV, respectively (Fig. 3c). The specific CB and VB potentials of g-C₃N₄ and 7%-BaCN-C₃N₄ are shown in Fig. 3d. Compared with g-C₃N₄, the band structure of 7%-BaCN-C₃N₄ moves downward, which might be attributed to the regulation of the band structure by Ba²⁺ and cyano group.

3.3. DFT calculations

How Ba²⁺ and cyano group affect the energy band and electronic structure of g-C₃N₄ are further discussed through comparing the DFT calculation results of the three models of g-C₃N₄, Ba-C₃N₄ (only doped

Ba²⁺) and BaCN-C₃N₄ (with Ba²⁺ doping and cyano defects). The calculation results show that Ba²⁺ could successfully inserted into the heptazine ring cavity of g-C₃N₄ (the schematic diagram is shown in Fig. 4d). In Fig. 4b, e, and h, the calculated band gaps of g-C₃N₄, Ba-C₃N₄ and BaCN-C₃N₄ are 1.84 eV, 0.89 eV and 0.2 eV, respectively, which are lower than the experimental values due to the known limitations of DFT calculation [41]. Obviously, the band gap of Ba-C₃N₄ is narrower than that of g-C₃N₄, indicating that the doping of Ba²⁺ narrows the band gap of g-C₃N₄. Compared with the other two models, the BaCN-C₃N₄ has the narrowest energy band, which proves that the cyano group can further optimize the energy band of g-C₃N₄. The narrower energy band, which is consistent with the conclusion of UV-vis DRS, is conducive to the absorption of visible light and the separation of photogenerated carriers. The theoretical calculation of the partial density of states (PDOS) shows that the minimum conduction band (CB) and maximum valence band (VB) of all catalysts are contributed by C 2p and N 2p, meanwhile Ba does not participate in the construction of band edge structure (in Fig. 4c, f, i). In Figure S3a-b, the repulsion of N lone electron pairs around the cavity makes the g-C₃N₄ appear corrugated after structural optimization, and the distance along the z axis is about 2.67 Å, which is similar to the conclusion of Azofra et al. [42]. Compared with g-C₃N₄, the corrugation amplitude of Ba-C₃N₄ is reduced by about 0.62 Å, which indicates that the narrowing of the band gap of g-C₃N₄ might be due to the optimization of the corrugated structure caused by the incorporation of Ba²⁺. In addition, the differential charge distribution further shows that Ba²⁺ are conducive to the enrichment of photogenerated electrons and improves the efficiency of photogenerated electron-hole separation (in Figure S3c). Moreover, as a strong electron withdrawing group, cyano group has also been proved to be beneficial to the transfer of photogenerated electrons [20].

3.4. Hydrogen production, pollutant degradation and intermediate products

Under visible light radiation, the photoreduction hydrogen generation efficiency of m-BaCN-C₃N₄ is investigated using H₂PtCl₆ as co-

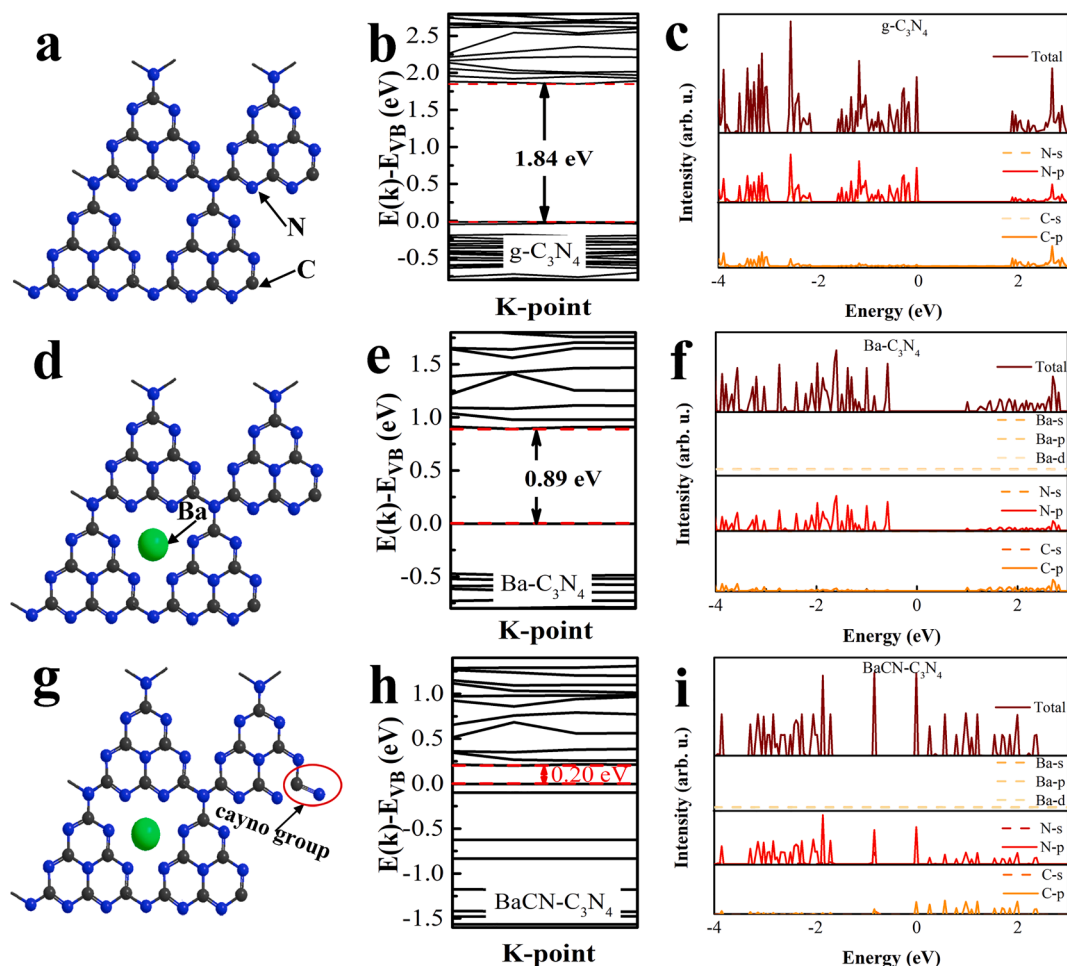


Fig. 4. Structural model (a, d, g), calculated band structures (b, e, h) and PDOS spectra (c, f, i) of g-C₃N₄, Ba-C₃N₄ and BaCN-C₃N₄.

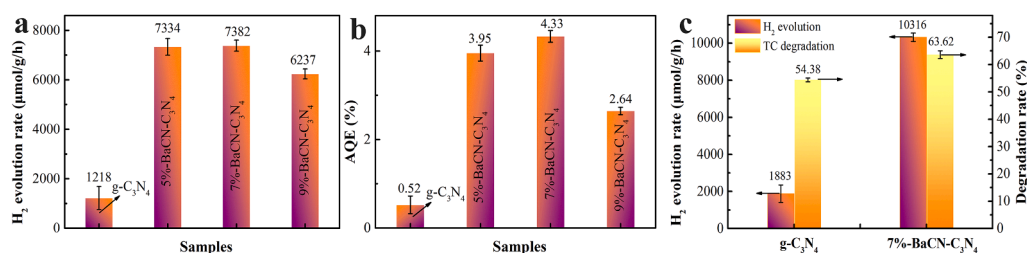


Fig. 5. Photocatalytic hydrogen evolution rate (a) and quantum yield (b) on g-C₃N₄ and BaCN-C₃N₄ photocatalysts under visible light ($\lambda > 420$ nm); synchronous degradation efficiency and hydrogen generation efficiency in TC wastewater (c).

catalyst and TEOA as scavenger. In Fig. 5a, the H₂ evolution rate of g-C₃N₄ is only 1218 μmol/g/h, which is lower than 5%-BaCN-C₃N₄ (7334 μmol/g/h), 7%-BaCN-C₃N₄ (7382 μmol/g/h) and 9%-BaCN-C₃N₄ (6237 μmol/g/h). With the introduction of Ba²⁺ and cyano group, the production of H₂ is increased and reached the maximum at 7%-BaCN-C₃N₄ (the yield is 6.10 times that of g-C₃N₄). However, the decrease in the hydrogen release rate of 9%-BaCN-C₃N₄ might be due to the decrease in its crystallinity which will lead to excessively defects, and some defects tend to form electron-hole pairs recombination centers [43]. Further, the quantum yield of all catalysts calculated by formula (1) is showed in Fig. 5b. The quantum yield of 7%-BaCN-C₃N₄ reached 4.33%, which has a good quantum conversion efficiency.

The simultaneous implementation of environmental pollution control and clean energy development has been studied by directly using waste water to produce hydrogen. In Fig. 5c, 20 mg/L tetracycline

wastewater replaces deionized water to study the activity of 7%-BaCN-C₃N₄ to remove environmental pollutants and hydrogen production at the same time. The experimental conclusions show that 7%-BaCN-C₃N₄ can achieve synchronous purification of TC and hydrogen generation, with efficiencies up to 63.62% and 10316 μmol/g/h, respectively. Compared with the deionized water solution, the hydrogen production efficiency in the TC solution is higher. This might be due to the fact that TC also acts as a holes scavenger, leading to more photogenerated electrons in the system participate in photoreduction to release hydrogen, while TC is oxidized and degraded by holes.

In addition, the system containing only TC and catalyst was used to study the oxidative degradation ability of BaCN-C₃N₄ to TC. The quasi-first-order equation $\ln(C_0/C_t) = kt$ is used to analyze the photocatalytic reaction kinetics, where C_0 and C_t are the TC concentration before visible light radiation and at the radiation time t , respectively, and k is

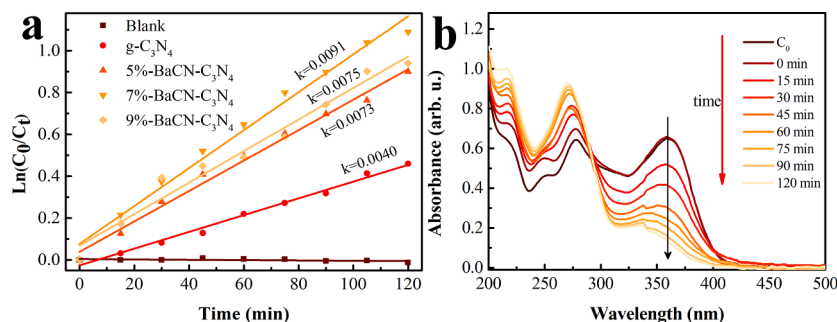


Fig. 6. Degradation kinetics of TC treated by BaCN-C₃N₄ alone (a) and UV-Vis absorption spectra of TC on 7%-BaCN-C₃N₄ at different times (b).

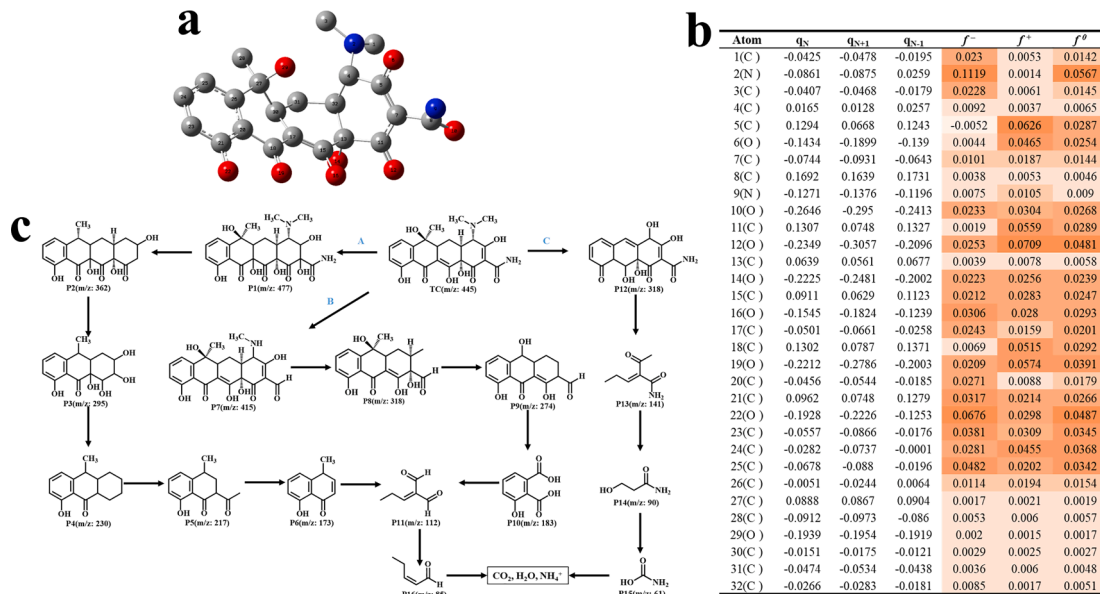


Fig. 7. The optimized structure of TC (gray, carbon; red, oxygen; blue, nitrogen) (a). The Hirshfeld charge distribution and Fukui index of TC (b). Possible degradation pathway of TC through 7%-BaCN-C₃N₄ photocatalyst (c).

the rate constant. In Fig. 6a, the k value of BaCN-C₃N₄ degradation of TC has been increased compared with g-C₃N₄. Among them, 7%-BaCN-C₃N₄ showed the highest k value (0.0091 min⁻¹), which is about 2.3 times that of g-C₃N₄. This might be closely related to the conversion of photo-generated electron-hole of 7%-BaCN-C₃N₄ into more strong oxidative substances. In Fig. 6b, the absorbance of TC ($\lambda_{\max} = 358$ nm) gradually decreases with the increase of the illumination time in the presence of 7%-BaCN-C₃N₄, which indicate that TC is degraded and oxidized. In addition, the absorption peak at 278 nm has a blue shift and the intensity gradually increases, the peak about 250 nm disappears and the intensity of the absorption peak at 215 nm gradually increases. These phenomena indicate that TC might be decomposed into other intermediate products

as the photocatalytic degradation progresses.

LC/MS analysis and Fukui index based on DFT on Gaussian 09 program are employed to speculate the reaction pathway of 7%-BaCN-C₃N₄ photocatalytic degradation of TC (detailed information is in Text S2). The intermediate obtained from the LC/MS analysis is shown in Figure S4. Before the reaction, TC is defined at 445 m/z due to its relative molecular weight. The geometrically optimized structure of TC molecule and the Fukui index (f^0) representing the radical attack of atoms on TC are shown in Fig. 7a-b. As shown, in TC molecules N2, C5, O6, C15, C17, C23, C21, O10, O12 and O16 atoms have larger f^0 values, which indicate that they are easily attacked by free radicals. Based on the detected intermediates and theoretical calculations, a possible TC

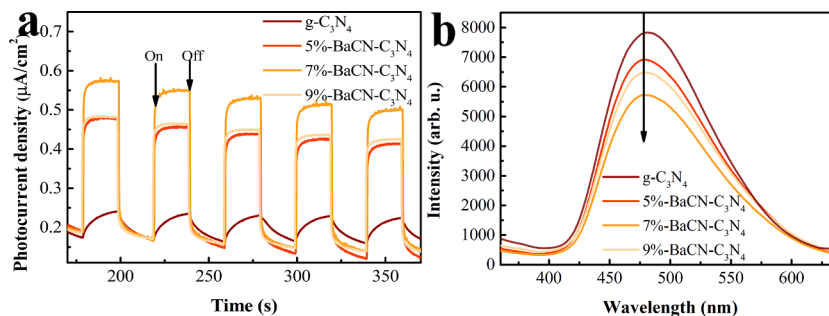


Fig. 8. Transient photocurrent responses (a) and PL spectra (b) of CN and m-BaCN-C₃N₄.

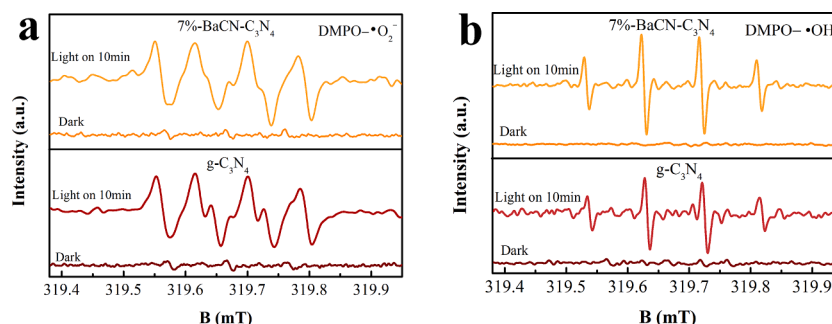


Fig. 9. DMPO ESR spectra were captured in methanol solution to $\bullet\text{O}_2^-$ (a) and deionized water to $\bullet\text{OH}$ (b). They were carried out in the dark and visible light ($\lambda > 420 \text{ nm}$) for 10 min.

degradation pathway is proposed in Fig. 7c. In the TC molecule, C5, C15 and C17 are easily hydroxylated by $\bullet\text{OH}$ due to their larger f^0 , resulting in the by-product P1 of m/z 477 [44]. The intermediate product P7 of m/z 415 is also observed, due to the loss of a methyl group at the N4 position and the amino group of the amide group at the C7 position [45]. Since free radicals are more likely to attack C21, C23, C24 and C25, the intermediate product P12 with m/z 318 is easily produced. Further, P1, P7 and P12 are decomposed into other intermediate products under the action of free radicals.

3.5. Charge separation and active species

The recombination of photogenerated carriers is another barrier which hinders the enhancement of photocatalytic activity [46,47]. The photoelectrochemical tests of $\text{g-C}_3\text{N}_4$ and $\text{m-BaCN-C}_3\text{N}_4$ revealed the excitation and separation of photogenerated carriers. In Fig. 8a, the corresponding intensity of the $\text{m-BaCN-C}_3\text{N}_4$ photocurrent time curve is higher than that of $\text{g-C}_3\text{N}_4$ (a typical on-off period under visible light irradiation). Among them, 7%-BaCN- C_3N_4 has the strongest photocurrent response intensity, which indicates that its photogenerated electron-hole separation efficiency is the best. Moreover, PL spectroscopy was used to further analyze the photo-generated electron-hole recombination efficiency. Fig. 8b shows that the PL emission spectra of $\text{g-C}_3\text{N}_4$, 5%-BaCN- C_3N_4 , 7%-BaCN- C_3N_4 and 9%-BaCN- C_3N_4 have an excitation peak at 452 nm [48]. Clearly, the intensity of 7%-BaCN- C_3N_4 is weaker than that of $\text{g-C}_3\text{N}_4$, which further shows that the recombination efficiency of photo-generated electron-hole pairs is lower. The above conclusions are matched with the DFT calculations, confirming that Ba^{2+} doping and cyano defects optimize the electronic structure of $\text{g-C}_3\text{N}_4$, thereby promoting the effective separation of photogenerated carriers of BaCN- C_3N_4 .

Through the trapping experiment, the main active species in the photocatalytic degradation TC system of $\text{g-C}_3\text{N}_4$ and 7%-BaCN- C_3N_4 were obtained. In the previously reported literature, the existence of h^+ , $\bullet\text{OH}$ and $\bullet\text{O}_2^-$ in the system can be confirmed by EDTA-2Na, BQ and IPA [49]. In Figure S6, the photocatalytic degradation of TC activity of the catalyst is inhibited after adding 1 mM of the corresponding capture agent, indicating that these active species play an important role in the TC degradation process. Among them, BQ and IPA have the greatest impact on the activity of the catalyst to degrade TC, which indicates that the main active substances for purifying TC are $\bullet\text{O}_2^-$ and $\bullet\text{OH}$. The determination of the active substance could further investigate the utilization of photo-generated electron-hole pairs, so the ESR spectra of $\text{g-C}_3\text{N}_4$ and 7%-BaCN- C_3N_4 are provided in Fig. 9. Under the excitation of visible light, the electrons in the valence band of $\text{g-C}_3\text{N}_4$ transition to CB to produce photo-generated electron-hole pairs. Electrons react with dissolved oxygen in water to generate superoxide radicals ($\bullet\text{O}_2^-$) ($E^0(\text{O}_2/\bullet\text{O}_2^-) = -0.33 \text{ eV vs. NHE}$) [50]. Electrons can also reduce O_2 to generate H_2O_2 ($E^0(\text{O}_2/\text{H}_2\text{O}_2) = 0.70 \text{ eV vs. NHE}$), and e^- further reacts with H_2O_2 to form $\bullet\text{OH}$ ($E^0(\text{H}_2\text{O}_2/\bullet\text{OH}) = 0.32 \text{ eV vs. NHE}$) [51,52]. The

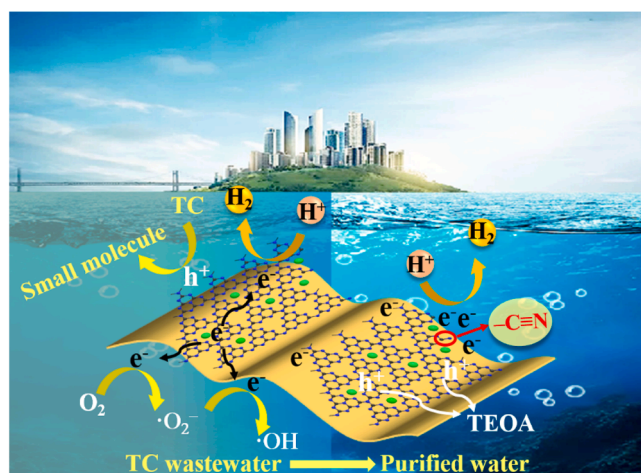


Fig. 10. Possible mechanism of BaCN- C_3N_4 simultaneous photocatalytic degradation of TC and hydrogen production (Note: the black ball represents the C atom, the blue ball represents the N atom, and the green ball represents the Ba ion).

$\text{g-C}_3\text{N}_4$ and 7%-BaCN- C_3N_4 catalyst systems have both $\bullet\text{O}_2^-$ and $\bullet\text{OH}$, but the 7%-BaCN- C_3N_4 catalyst system has a higher $\bullet\text{OH}$ content compared to $\text{g-C}_3\text{N}_4$. Because $\bullet\text{OH}$ has a stronger oxidation capacity than $\bullet\text{O}_2^-$, 7%-BaCN- C_3N_4 could degrade TC more efficiently, and it is further confirmed that the photo-generated electron-hole pairs in 7%-BaCN- C_3N_4 are well separated and utilized.

3.6. Photocatalytic mechanism

Based on experimental and DFT calculations results, a reasonable mechanism is proposed that Ba^{2+} and cyano group synergistically enhance the photocatalytic performance of $\text{g-C}_3\text{N}_4$ for hydrogen production in wastewater. The synergistic effect of Ba^{2+} doping and cyano defects narrows the band gap of BaCN- C_3N_4 , which is more conducive to visible light absorption. Moreover, the photogenerated carriers might be quickly separated by the enriched electron effect of cation Ba^{2+} and cyano group. In three different systems, namely hydrogen evolution, hydrogen evolution in TC and degradation of TC, the photocatalytic mechanism of BaCN- C_3N_4 is discussed separately. 1) hydrogen evolution (no oxygen environment). The photogenerated electrons reduce the H^+ into hydrogen ($2\text{H}^+ + 2\text{e}^- = \text{H}_2$, $E_0 = 0 \text{ eV vs. NHE}$), while the photo-generated holes are consumed by the electron donor TEOA, thereby realizing the reaction of water splitting to produce hydrogen. 2) Hydrogen generation in TC wastewater (no oxygen environment). The macromolecule TC and the small molecule TEOA are used together as the electron donor, which means that the photo-generated holes are consumed by more electron donors, and more photo-generated electrons

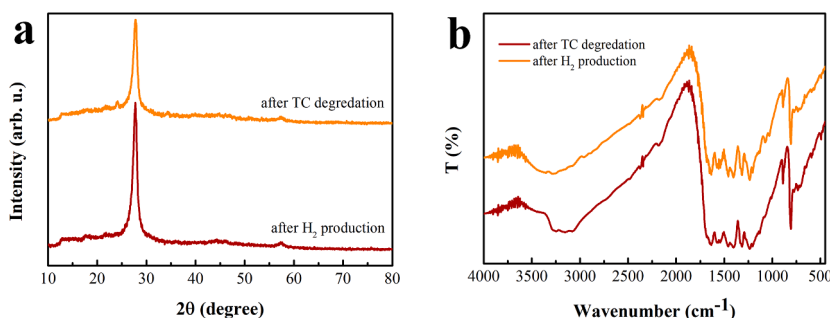


Fig. 11. Comparison of XRD (a) and FT-IR (b) before and after photo-oxidation of TC and photo-reduction of H₂ on 7%-BaCN-C₃N₄.

are produced by reducing hydrogen. Moreover, BaCN-C₃N₄ has a more positive valence band position than g-C₃N₄, implying that it has a stronger oxidation properties to purify TC. 3) Photocatalytic degradation of TC (open environment). The photo-generated holes excited by the BaCN-C₃N₄ catalyst participate in the oxidation of TC. Then, the photogenerated electrons are further used for the oxidation of TC by reducing the dissolved oxygen and other substances in the water to generate free radicals (e.g. •O₂⁻, •OH) with stronger oxidizing ability (these free radicals are confirmed by ESR spectroscopy). In Fig. 10, a possible mechanism for BaCN-C₃N₄ photocatalytic degradation of organic wastewater under visible light to obtain pure water and hydrogen is proposed.

3.7. Recyclability and stability

The stability of photocatalyst is also a major factor limiting its application. Therefore, we conducted four cycle tests to investigate the stability of 7%-BaCN-C₃N₄ photocatalytic degradation of TC and hydrogen production (in Text S4). In Figure S7a-b, the 7%-BaCN-C₃N₄ photocatalyst could still maintain high photocatalytic activity after repeated 4 times in new energy development and pollutant treatment. The XRD and FT-IR spectra before and after use show that the structure of 7%-BaCN-C₃N₄ has not changed significantly, and the weakening of the reproducibility effect may be due to the reduction of active sites rather than structural damage (Fig. 11). The above results show that BaCN-C₃N₄ could be used as both “source control” (evolution of clean energy) and “end treatment” (removal of pollutants) and has outstanding stability, so it has broad application prospects.

4. Conclusions

In short, we obtained the Ba²⁺ doping and cyano defects co-modified new bifunctional catalyst BaCN-C₃N₄ by a clever salt-assist method, which can simultaneously degrade tetracycline to ameliorate the environment and decompose water to obtain hydrogen energy. In the same system for contaminant control and hydrogen generation, the TC degradation efficiency and hydrogen production rate are about 1.20 and 5.50 times than that of pristine g-C₃N₄, respectively, and it has excellent recycling performance. In the BaCN-C₃N₄ structure, the doped Ba²⁺ promotes the opening of the heptazine ring to produce cyano defects. Furthermore, the synergistic effect of Ba²⁺ and cyano group narrows the band gap of g-C₃N₄, improves the utilization of visible light and promotes the separation of photogenerated electron-hole pairs. This research demonstrates that g-C₃N₄ has great potential in photocatalytic applications and paves the way for a new g-C₃N₄-based photocatalyst that could concurrently complete water remediation and H₂ evolution.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2021.130278>.

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